Justin Palmer

Software Engineering / Cloud & Data Architecture

justinraypalmer.com/contact

PROFESSIONAL PROFILE

Senior Cloud Engineer and Data Architect with over 16 years of experience delivering secure, enterprise-scale data solutions. Specialized in architecting Azure-native Big Data pipelines, governed analytics, and Generative AI solutions that democratize data access across complex enterprise domains. Expert in AI-accelerated delivery, orchestrating advanced agentic workflows (Windsurf, Codex, ChatGPT Pro) to transform development velocity and ship production-ready applications with superior code integrity. Proven track record of designing secure network architectures and API ecosystems in high-compliance federal sectors.

TECHNICAL SKILLS

Cloud Data Architecture: Azure Synapse (Dedicated Pools), Azure Data Factory, Azure SQL, Data Lake Gen2, Event Grid, Blob Storage.

Full-Stack Engineering: C#, ASP.NET Core, Blazor, React, Node.js, Python, REST API Design, Swagger/OpenAPI.

AI & Automation: Azure OpenAI Service, RAG Patterns, AI-Augmented Development (Windsurf, Codex), PowerShell Automation.

Security & Networking: Azure VNETs, Private Endpoints, Inbound/Outbound Traffic Security, Vulnerability Management (Nexus), OAuth/OIDC.

DevOps & Governance: Azure DevOps (CI/CD), Infrastructure as Code (IaC), Row-Level Security (RLS), Data Governance.

PROFESSIONAL EXPERIENCE

Senior Cloud Engineer / Technical Lead

U.S. Department of the Treasury | Nov 2022 - Present

- Architected and integrated a custom Azure OpenAI solution using a RAG (Retrieval-Augmented Generation) pattern within the enterprise Blazor portal. Engineered the system to index and synthesize millions of Power BI audit log events from the Data Lake, empowering administrators to diagnose performance bottlenecks via natural language queries.
- Serve as a Lead Cloud Architect for enterprise Big Data processing utilizing Azure Synapse Dedicated Pools and Azure SQL. Design robust low-code ETL pipelines in Data Factory complemented by high-code Azure Functions in .NET, Node, or Python for complex data transformations.
- Design and secure the complete network infrastructure for enterprise applications, configuring advanced ingress/egress controls, private endpoints, and API gateways to ensure strict compliance with federal security standards.
- Develop and maintain the agency’s core API ecosystem, utilizing ASP.NET Core and ensuring comprehensive documentation and discoverability via Swagger/OpenAPI.
- Leverage AI-augmented scripting (PowerShell) to automate security operations, including a proactive system that integrates with Nexus scans to detect software vulnerabilities and immediately quarantine compromised Virtual Machines.
- Architected a massive Human Capital Tabular Model (25 fact tables, 90 dimensions) tracking workforce actions and pay. Engineered advanced optimization strategies, including incremental refresh policies and programmatic partition management, to ensure sub-second query performance.
- Developed a centralized Blazor Web App acting as an administrative hub to monitor platform health and developer velocity, providing granular visibility into user access and ensuring strict compliance.

PROFESSIONAL EXPERIENCE / CONTINUED

Senior Data Engineer / Analyst

Internal Revenue Service | Jun 2019 - Nov 2022

- Managed the transition of training analytics to the Treasury’s enterprise warehouse, consolidating 20+ legacy PL/SQL scripts into a unified Big Data model processing 20M+ rows daily.
- Developed .NET/SQL reconciliation utilities to synchronize data across three disparate HR/payroll systems, identifying architectural flaws and improving institutional data integrity.
- Engineered automated data pipelines using Python and APIs (CDC/JHU) to ingest, transform, and publish critical COVID-19 tracking data with zero manual intervention.
- Designed signature HR dashboards serving 500k+ views to 500+ stakeholders, reducing ad-hoc data requests by democratizing access to self-service analytics.

Operations Research Analyst

Internal Revenue Service | Nov 2017 - Jun 2019

- Replaced a $3M legacy organizational modeling tool with a custom, governed solution, encoding complex hierarchy rules and driving enterprise adoption.
- Implemented R-based outlier detection and forecasting models to automate the ingestion and analysis of complex paginated records.

EDUCATION

Master of Business Administration (MBA) | Finance & Analytics - Murray State University (4.0 GPA)

Bachelor of Science (BS) | Accounting - University of Kentucky

CERTIFICATIONS

Microsoft Certified: Azure AI Fundamentals (AI-900)

Microsoft Certified: Azure Data Fundamentals (DP-900)

Microsoft Certified: Azure Fundamentals (AZ-900)